

MODELS	GSM8K (PEARSON = 0.93)		ECQA (PEARSON = 0.98)	
	$\mathbf{T}_{\text{Accuracy}}(\%)$	$\mathbf{A}_{\text{Accuracy}}(\%)$	$\mathbf{T}_{\text{Accuracy}}(\%)$	$\mathbf{A}_{\text{Accuracy}}(\%)$
LLama-7B	36.48 $\pm$ 0.92	68.41 $\pm$ 0.32	70.40 $\pm$ 0.92	61.62 $\pm$ 0.01
LLama2-7B	40.71 $\pm$ 0.16	71.22 $\pm$ 0.12	72.34 $\pm$ 0.22	61.96 $\pm$ 0.02
LLama-13B	42.07 $\pm$ 0.15	72.25 $\pm$ 0.23	72.74 $\pm$ 0.43	61.89 $\pm$ 0.01
LLama2-13B	47.29 $\pm$ 1.24	73.06 $\pm$ 0.78	74.76 $\pm$ 0.56	62.29 $\pm$ 0.01

Table 1: The final task accuracy ($\mathbf{T}_{\text{Accuracy}}$) and the assessment accuracy ($\mathbf{A}_{\text{Accuracy}}$) of different vanilla fine-tuned models. $\mathbf{T}_{\text{Accuracy}}$ and $\mathbf{A}_{\text{Accuracy}}$ exhibit a strong positive correlation, with Pearson Correlation Coefficients of 0.93 and 0.98 at GSM8K and ECQA, respectively.

induce to the incorrect answer, respectively. Upon manually examining 50 examples, we observe that the quality of positive COTs is noticeably better than that of negative COTs.

We further compute the token-averaged log-likelihood score of each positive and negative COT c using the fine-tuned LLMs as follows:

$$s_{\theta}^c = \frac{1}{|c|} \sum_{j=1}^{|c|} \log P(c_j \mid c_{<j}, q; \theta), \quad (2)$$

where q is the corresponding question. It is reasonable to expect that the fine-tuned LLMs will be able to assess the quality of different candidate COTs of previously encountered questions, i.e., assigning higher scores to the positive ones. Therefore, we use an assessment accuracy $\mathbf{A}_{\text{Accuracy}}$ to assess the capability of fine-tuned LLMs in assigning appropriate scores to various COTs:

$$\mathbf{A}_{\text{Accuracy}} = \frac{1}{9N} \sum_{i=1}^N \sum_{j=1}^3 \sum_{k=1}^3 \mathbb{I}(s_{\theta}^{c_i^{pj}} > s_{\theta}^{c_i^{nk}}) \quad (3)$$

As shown in Table 1, the assessment accuracy of the VFT-LLMs falls short of expectations, with an average $\mathbf{A}_{\text{Accuracy}}$ of merely around 70% on GSM8K and 62% on ECQA, respectively. Note that this is a two-class classification problem where a random baseline can achieve the 50.00% accuracy. These results show that the assessment ability of VFT-LLMs is far from expected, as they cannot accurately discern the quality of various COTs of previously learned questions. This behavior of VFT-LLMs is not consistent with that of humans, as humans have the ability to access the quality of different COTs after learning to reason. In addition, we also notice that LLMs with stronger reasoning abilities have better assessment accuracy. Specifically, the task accuracy and the assessment accuracy exhibit a strong positive correlation, with Pearson Correlation Coefficients of 0.93 and 0.98 at GSM8K and ECQA, respectively. This observation inspires us to improve the reasoning ability of LLMs by aligning their scoring behaviors with the golden standard assessment.

4 METHODOLOGY

We have demonstrated that the scoring behaviors of vanilla fine-tuned LLMs exhibit misalignment with the gold standard assessment. In this section, we propose an alignment fine-tuning (AFT) paradigm to address this problem to enhance their reasoning ability. Specifically, on top the VFT objective $\mathcal{L}_{VFT}$, AFT further introduce an alignment objective $\mathcal{L}_A^*$:

$$\mathcal{L}_{AFT} = \mathcal{L}_{VFT} + \mathcal{L}_A^*. \quad (4)$$

In the following part of this section, we will introduce the design process of $\mathcal{L}_A^*$.

4.1 GENERATE COTs FOR TRAINING DATA

To implement AFT, we first need to generate multiple COTs for each question in the training set. For each training example (q, c, a) , we first sample k generation results $\{(c_i, a_i)\}_{i=1}^k$ from the VFT-LLMs based on the input question q . Then, we divide these generation results into two groups, namely positive group $\mathbf{G}_P$ and negative group $\mathbf{G}_N$, based on the correctness of their answer. Formally, a generation results (c_i, a_i) belongs to $\mathbf{G}_P$ if $a_i = a$, otherwise it is part of $\mathbf{G}_N$. Generally, the quality of COTs in the positive group $\mathbf{G}_P$ is better than that of $\mathbf{G}_N$.

4.2 ALIGNMENT

As demonstrated by our pilot experiment, VFT-LLMs fail to give reasonable scores to COTs in $\mathbf{G_P}$ and $\mathbf{G_N}$. To align the scoring behaviors of LLMs with the golden standard assessment, we need to design an objective to let the scores of all positive COTs in $\mathbf{G_P}$ larger than that of negative COTs in $\mathbf{G_N}$. This objective bears resemblance to contrastive learning, which aims to ensure that the score of positive example is larger than those of all negative examples, utilizing an InfoNCE loss:

$$\mathcal{L}_{InfoNCE} = -\log \left[\frac{\exp(s_{\theta}^{c_p})}{\exp(s_{\theta}^{c_p}) + \sum_{c_n \in \mathbf{G_N}} \exp(s_{\theta}^{c_n})} \right] = \log \left[1 + \sum_{c_n \in \mathbf{G_N}} \exp(s_{\theta}^{c_n} - s_{\theta}^{c_p}) \right] \quad (5)$$

Intuitively, minimizing Equation 5 aims to make the positive score $s_{\theta}^{c_p}$ larger than all negative scores. However, since there is more than one positive example in $\mathbf{G_P}$, inspired by (Su et al., 2022; Wang et al., 2022), we extend $\mathcal{L}_{InfoNCE}$ to accommodate multiple positive examples:

$$\mathcal{L}_A = \log \left[1 + \sum_{c_p \in \mathbf{G_P}} \sum_{c_n \in \mathbf{G_N}} \underbrace{\exp(s_{\theta}^{c_n} - s_{\theta}^{c_p})}_{\text{alignment term}} \right] \quad (6)$$

where s_{θ}^c is the average log-likelihood score of the COT c calculated by Equation 2. Minimizing $\mathcal{L}_A$ encourages all positive scores to be larger than all negative scores.

4.3 CONSTRAINT

Nevertheless, although the quality of negative COTs may not be as high as that of positive COTs, they still retain a respectable quality, as they are sampled from fine-tuned, powerful LLMs. We find that reducing their scores by Equation 6 without setting any constraint will result in the degradation of the LLMs. Therefore, we further design two constrained methods, Detached Constraint (DC), and Boundary Constraint (BC) to avoid such catastrophe.

4.3.1 DETACHED CONSTRAINT

To prevent model degradation, DC adds constraint to negative scores by detaching their gradient:

$$\mathcal{L}_A^{DC} = \log \left[1 + \sum_{c_p \in \mathbf{G_P}} \sum_{c_n \in \mathbf{G_N}} \underbrace{\exp(\mathbf{D}(s_{\theta}^{c_n}) - s_{\theta}^{c_p})}_{\text{detached alignment term}} \right], \quad (7)$$

where $\mathbf{D}(\cdot)$ denotes the detach operation, which means the gradient would not back-prop through the negative scores. As a results, $\mathcal{L}_A^{DC}$ achieves the alignment by only increasing positive scores without explicitly decreasing negative ones.

4.3.2 BOUNDARY CONSTRAINT

Besides DC, we also want to explore whether better results can be obtained by marginally decreasing negative scores. To this end, we propose BC that adds a constraint term to $\mathcal{L}_A$:

$$\mathcal{L}_A^{BC} = \log \left\{ 1 + \sum_{c_p \in \mathbf{G_P}} \sum_{c_n \in \mathbf{G_N}} \left[\underbrace{\exp(s_{\theta}^{c_n} - s_{\theta}^{c_p})}_{\text{alignment term}} + \underbrace{\exp(T - s_{\theta}^{c_n})}_{\text{constraint term}} \right] \right\} \quad (8)$$

Intuitively, the constraint term increases the score of the negative COT $s_{\theta}^{c_n}$, with the extent of improvement regulated by the value of T . We aim for T to achieve the effect of increasing $s_{\theta}^{c_n}$ when it is lower than a boundary B . In this paper, we chose B as the minimum positive COT score minus a hyper-parameter β , i.e., $B = s_{\theta}^{c_p^*} - \beta$, where $s_{\theta}^{c_p^*} = \min_{c_p \in \mathbf{G_P}} s_{\theta}^{c_p}$. To achieve this, we analyze the

gradient of Equation 8 with respect to the parameters θ :

$$\begin{aligned} \nabla_{\theta} \mathcal{L}_A^{BC} &\propto - \sum_{c_p \in \mathbf{G_P}} \sum_{c_n \in \mathbf{G_N}} [\exp(s_{\theta}^{c_n} - s_{\theta}^{c_p})(\nabla_{\theta} s_{\theta}^{c_p} - \nabla_{\theta} s_{\theta}^{c_n}) + \exp(T - s_{\theta}^{c_n}) \nabla_{\theta} s_{\theta}^{c_n}] \\ &= - \sum_{c_p \in \mathbf{G_P}} \sum_{c_n \in \mathbf{G_N}} \left\{ \underbrace{\exp(s_{\theta}^{c_n} - s_{\theta}^{c_p}) \nabla_{\theta} s_{\theta}^{c_p}}_{\text{increase } s_{\theta}^{c_p}} + \underbrace{[\exp(T - s_{\theta}^{c_n}) - \exp(s_{\theta}^{c_n} - s_{\theta}^{c_p})] \nabla_{\theta} s_{\theta}^{c_n}}_{\text{change } s_{\theta}^{c_n} \text{ based on the coefficient}} \right\} \end{aligned} \quad (9)$$

Because the score $s_{\theta}^{c^*}$ increases along the gradient $\nabla_{\theta} s_{\theta}^{c^*}$, based on $\nabla_{\theta} \mathcal{L}_A^{BC}$, for each pair (c_p, c_n) , $\mathcal{L}_A^{BC}$ consistently increases $s_{\theta}^{c_p}$ due to the positive coefficient $\exp(s_{\theta}^{c_n} - s_{\theta}^{c_p}) > 0$. Additionally, it elevates the negative score $s_{\theta}^{c_n}$ when:

$$\exp(s_{\theta}^{c_n} - s_{\theta}^{c_p}) < \exp(T - s_{\theta}^{c_n}) \Rightarrow s_{\theta}^{c_n} < \frac{T + s_{\theta}^{c_p}}{2} = B \quad (10)$$

Otherwise, it tends to decrease or keep the score of $s_{\theta}^{c_n}$. Combing $B = s_{\theta}^{c_p^*} - \beta$ and Equation 10, we can achieve the value of $T = 2s_{\theta}^{c_p^*} - 2\beta - s_{\theta}^{c_p}$.

4.4 EXTENDING TO RANKING ALIGNMENT

The quality of the different COTs is not a simple binary relationship $\mathbf{G_P} \succ \mathbf{G_N}$, i.e., the quality of positive COTs is better than that of negative COTs. In a more general situation, COTs in each group can also have quality differences, which means the quality of all generated COTs can be ranked as a sequence $c_1 \succeq c_2 \succeq \dots \succeq c_k$. If we can obtain such a quality ranking sequence, we can easily extend our binary-feedback boundary constraint alignment loss $\mathcal{L}_A^{BC}$ to a ranking-feedback boundary-constrained alignment loss as follows:

$$L_A^{RBC} = \log \left\{ 1 + \sum_{c_i \succ c_j} \left[\underbrace{\exp(s_{\theta}^{c_j} - s_{\theta}^{c_i})}_{\text{alignment term}} + \underbrace{\exp(2s_{\theta}^{c_j^*} - 2\beta - s_{\theta}^{c_i} - s_{\theta}^{c_j})}_{\text{constraint term}} \right] \right\} \quad (11)$$

Where $s_{\theta}^{c_j^*} = \min_{c_k \succ c_j} s_{\theta}^{c_k}$ is the minimal score of COTs that have the better quality than c_j . Compared with $\mathcal{L}_A^{BC}$, L_A^{RBC} can bring LLMs more detailed training signals of the COT assessment, which can further enhance their performance. We also try to extend $\mathcal{L}_A^{DC}$ to the ranking situation, and we find it slightly underperforms in comparison to L_A^{RBC} . Please refer to Appendix C for details.

5 EXPERIMENTS

5.1 EXPERIMENTAL SETUPS

Datasets We conduct our experiments on three widely used reasoning datasets with human-annotated chain-of-thoughts, including math reasoning tasks **GSM8K** (Cobbe et al., 2021), **AQUA-RAT** (Ling et al., 2017), commonsense reasoning task **ECQA** (Aggarwal et al., 2021). Furthermore, we create **GSM8K-RANK** to evaluate the effectiveness of our AFT in the ranking situation. Please refer to Appendix A for more details of these datasets.

Parameter Setting We conduct experiments on four large language models, LLama(2)-7B and LLama(2)-13B. We do not conduct experiments on larger models due to resource limitations. We sample $k = 6$ COTs from VFT-LLMs with a sampling temperature of 1. Our detached constraint alignment loss does not introduce any hyper-parameters, and we search the boundary constraint hyper-parameter β based on the validation set. For more training details, please refer to Appendix B.

Baselines We compare our AFT with the following baselines: 1) **VFT**: the vanilla fine-tuning (VFT) method that simply trains LLMs with the reference COT using the MLE loss, which is the most widely used training strategy; 2) **RFT**: Rejective sampling fine-tuning (RFT) (Yuan et al., 2023a) selects the COTs with the correct answer, adds these COTs to the origin training data, and uses the